ARCHITECTURE REFERENCE · SYSTEM VERSION 2026.2

🏗️ PillCare Platform Architecture & Integration Specification

Decoupled 2-Repository Microservice Ecosystem & Technical Contract Documentation

BACKEND ENDPOINT	PRIMARY WEB DOMAIN	ADMIN MISSION CONTROL
gcp-cloud-run / europe-west1	https://pillcare.in	https://admin.pillcare.in

1. Multi-Repository System Architecture

PillCare operates as a decoupled microservice architecture divided across two core repositories:

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┌──────────────────────────────────────────────────────────────────────┐ │ REPOSITORY 1:
Pillcare.com (Static Website & Admin Panel) │ │ - Landing Page: pillcare.in │ │ - Mission Control UI:
admin.html / admin/index.html │
└──────────────────────────────────────────────────────────────────────┘ │ HTTPS REST APIs ▼
┌──────────────────────────────────────────────────────────────────────┐ │ REPOSITORY 2: Pillcare-
Reminder_V01 (Mobile App & Core Backend API) │ │ - FastAPI Backend Server: backend/server.py │ │ -
Mobile Application: React Native (Expo SDK 50) │ │ - Database Layer: MongoDB Atlas (db.users,
db.subscriptions, etc.) │ │ - Push Gateway: Expo Push API (exp.host) │
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REPOSITORY COMPONENT	GITHUB ORIGIN	LOCAL SYSTEM PATH	SCOPE & RESPONSIBILITIES
Repo 1: Website & Admin	`ntnr737/Pillcare.com`	`.../pillcare-website`	Marketing landing pages, download links, `admin.html` executive dashboard UI.
Repo 2: App & Backend	`ntnr737/Pillcare-Reminder_V01`	`.../Pillcare-Reminder_V01`	FastAPI server (`server.py`), React Native cross-platform application (`frontend/`).

2. End-to-End Integration Contracts & Protocols

A. Subscription Tiers & 1-Click Feature Override Engine

Admin Panel updates subscription plans (Free, Basic ₹30, Premium ₹99, Enterprise) and billing cycles (Monthly 30d, Annual 365d) or toggles specific features (OCR Scanner, PDF Export, Yoga Full, Brand → Generic, Caregiver Alerts).

`POST` `/admin/users/{user_id}/subscription` — Set subscription plan, billing cycle, expiration, and notes.
`POST` `/admin/users/{user_id}/features` — Apply single-click feature overrides (`export_data` , `scanner` , `yoga_full` , etc.).
`DELETE` `/admin/users/{user_id}/features/{feature}` — Remove a manual feature override.
`GET` `/api/me/features` — Mobile client endpoint to sync active user privileges.

B. High-Reliability Push Notification Pipeline

Supports both `ExpoPushToken[...]` and `ExponentPushToken[...]` formats across Expo SDK versions.

`POST` `/api/profile/push-token` — Registers or updates device push token (accepts `{ push_token }` or `{ token }`).
`POST` `/admin/notify` — Broadcasts push notifications globally or to specific target `user_id` / `email`.

C. Flash Announcements & Healthcare Intelligence

Allows broadcasting instant in-app promotions, system alerts, or feature releases to specific subscription tiers.

`POST` `/admin/announcements` — Publish announcement (`title`, `body`, `type`, `target_plan`, `link_url`).
`GET` `/admin/analytics/financials` — Unit economics (MRR, ARR, ARPU, LTV, Paid Conversion %).
`GET` `/admin/analytics/healthcare-intelligence` — Refill GMV, Low-Stock Radar (≤3 doses remaining).

3. MongoDB Database Schema Definitions

COLLECTION	KEY FIELDS & INDEXES	PURPOSE & SYSTEM FUNCTION
`users`	`id`, `email`, `name`, `push_token`, `created_at`	Core user identity and device push token registration records.
`profile`	`user_id`, `phone`, `caregiver_phone`, `location_city`, `location_state`	Onboarded patient demographics and caregiver contact details.
`subscriptions`	`user_id`, `plan`, `billing_cycle`, `status`, `overrides`	Subscription status, billing cycle expirations, and custom feature flags.
`medications`	`id`, `user_id`, `name`, `dosage`, `frequency`, `stock`, `active`	Active and paused prescription items with current inventory stock.
`doses`	`id`, `user_id`, `medication_id`, `scheduled_time`, `status`	Adherence history (taken, missed, skipped) used for analytics.
`announcements`	`id`, `title`, `body`, `target_plan`, `created_at`	Active flash announcement banners displayed on mobile home screen.

4. Performance & Concurrency Rules

Concurrent Batch Aggregation (`asyncio.gather`): All admin endpoints returning user lists or analytics avoid sequential `for` loop queries, processing database queries concurrently to maintain sub-50ms latency. **FastAPI Static Route Order**: Static endpoints (e.g. `/admin/analytics/users/comprehensive`) are declared *before* parameterized endpoints (e.g. `/admin/analytics/users/{user_id}`) to prevent route matching collisions. **Authentication Enforcer**: `require_admin` dependency validates header `x-admin-secret` matching `ADMIN_SECRET` environment variable or valid Google OAuth admin tokens.

5. Recent System Enhancements (Version 2026.2 / Release v1.0.11)

🌐 100% Internationalization (i18n) QC

- **Languages Supported**: English (`en`), Hindi (`hi`), Bengali (`bn`), German (`de`), Spanish (`es`), French (`fr`).
- **Zero English Fallbacks**: 100% full translation dictionary coverage across all 6 main tabs and 6 modal components.
- **Dynamic Parameter Support**: `LanguageContext.tsx` supports dynamic variables (`{med}`, `{slot}`, `{count}`, `{avg}`).

🔐 Google OAuth 2.0 SHA-1 Setup

- **Package Name**: `com.pillcare.reminder`
- **SHA-1 Fingerprint**: `5E:8F:16:06:2E:A3:CD:2C:4A:0D:54:78:76:BA:A6:F3:8C:AB:F6:25`
- **Web Client ID**: `1082668880575-lfoqgnieodb52b5ges1vppiv92ifbioc.apps.googleusercontent.com`
- Fully aligned in Google Cloud Console for seamless native Google Sign-In on Android.

📦 Production Release Artifact & Website Sync

Current Production Release: `v1.0.11` (Tag: v1.0.11)
Android APK Binary Asset: PillCare-v1.0.11-release.apk (113 MB)
Website & Admin Sync: Landing page (`index.html`) and Admin Panel (`admin/index.html`, `admin.html`) updated to point directly to the release asset.

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